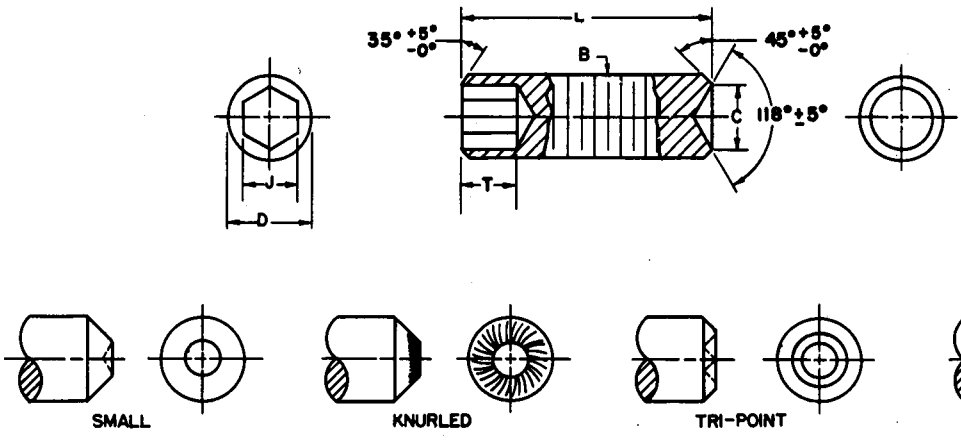
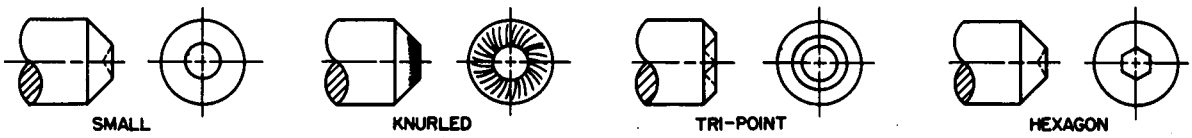
FED. SUP CLASS
5305





ACCEPTABLE CUP POINT DESIGNS

D	NOMINAL SIZE		#2 (.086)		#4 (.112)		#6 (.138)		#8 (.164)		#10 (.190)	
B	THREADS PER INCH		56NC-3A		40NC-3A		32NC-3A		32NC-3A		24NC-3A	
J	WIDTH OF SOCKET ACROSS FLAT	Max	.0355		.051		.0635		.0791		.0947	
		Min	.035		.050		.0625		.0781		.0937	
C	DIAMETER OF POINT	Max	.047		.061		.074		.087		.102	
		Min	.039		.051		.064		.076		.088	
T	KEY ENGAGEMENT	Min	.028		.040		.050		.062		.075	
L	LENGTH	Tolerance	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN
1/8			1		9		20		33			
3/16			2		10		21		34		46	
1/4		± 1-1/2%	3		11		22		35		47	
5/16		of L with a minimum			12		23		36		48	
3/8		of ± .010			13		24		37		49	
7/16		in.			14		25		38		50	
1/2					15		26		39		51	
5/8					16		27		40		52	
3/4							28		41		53	
7/8							29		42		54	
1							30		43		55	
1-1/4											56	

KEY ENGAGEMENT: These dimensions apply to screws one diameter in length or longer. For screws shorter than one diameter in length, socket to be as deep as practicable.
MATERIAL: Steel, High-Grade Alloy, in accordance with Procurement Specification.
HARDNESS: Setscrews shall be hardened by Heat Treatment to a hardness of Rockwell C45 to 53.
PROTECTIVE COATING: Phosphate coated, in accordance with MIL-C-16232, Type II.
THREADS: The threads shall be in accordance with Screw-Thread Standards for Federal Services, Handbook H-28.
DIMENSIONS: All dimensions are in inches unless otherwise specified.
PART NUMBER: The MS part number consists of the MS number, plus the dash number. Example: MS51019-1.
NOTES: (1) Referenced documents shall be of the issue in effect on the date of invitations for bid.
(2) This standard takes precedence over documents referenced herein.

Ⓐ CANCELLED
SEE SHEET 2

This document has been promulgated by the Department of Defense as the Military Standard to limit the selection of the item, product, or design covered herein in engineering, design and procurement. This standard shall become effective not later than 90 days after the latest date of approval shown.

Ⓐ For changes see sheets 1 and 2

APPROVED 7 MAY 58 REVISED Ⓐ 29 JANUARY 1964

P.A. WC Other Cust SHIPS ASD	TITLE SETSCREW, HEXAGON SOCKET, CUP POINT, ALLOY STEEL, PHOSPHATE COATED, NC-3A AND UNC-3A	MILITARY STANDARD MS51019	
PROCUREMENT SPECIFICATION FF-S-103	SUPERSEDES:	SHEET 1	OF 2

DD FORM 1 SEP 67 672-1 (Continued) PREVIOUS EDITION OF THIS FORM IS OBSOLETE.

NOTICE.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

Source: https://assist.dla.mil -- Downloaded: 20240-23T17:53Z
Check the source to verify that this is the current version before use.

FED. SUP CLASS 5305

D	NOMINAL SIZE		1/4		5/16		3/8		7/16		1/2
B	THREADS PER INCH		20UNC-3A		18UNC-3A		16UNC-3A		14UNC-3A		13UNC-3A
J	WIDTH OF SOCKET ACROSS FLAT	Max	.127		.1582		.1895		.2207		.252
		Min	.125		.1562		.1875		.2187		.250
C	DIAMETER OF POINT	Max	.132		.172		.212		.252		.291
		Min	.118		.156		.194		.232		.270
T	KEY ENGAGEMENT	Min	.100		.125		.150		.175		.200

L LENGTH	Tolerance	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN
3/16		62									
1/4		63		81		99					
5/16		64		82		100					
3/8	±1-1/2% of L with a minimum of ± .010 in.	65		83		101		118		135	
7/16		66		84		102		119		136	
1/2		67		85		103		120		137	
5/8		68		86		104		121		138	
3/4		69		87		105		122		139	
7/8		70		88		106		123		140	
1		71		89		107		124		141	
1-1/4		72		90		108		125		142	
1-1/2		73		91		109		126		143	
1-3/4		74		92		110		127		144	
2		75		93		111		128		145	
2-1/2						112		129		146	
3										147	

D	NOMINAL SIZE		5/8		3/4		7/8		1		
B	THREADS PER INCH		11UNC-3A		10UNC-3A		9UNC-3A		8UNC-3A		
J	WIDTH OF SOCKET ACROSS FLAT	Max	.3155		.378		.503		.5655		
		Min	.3125		.375		.500		.5625		
C	DIAMETER OF POINT	Max	.371		.450		.530		.609		
		Min	.347		.425		.502		.579		
T	KEY ENGAGEMENT	Min	.250		.300		.400		.450		

L LENGTH	Tolerance	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN	Dash No.	FIIN
1/2		153									
5/8		154		169							
3/4		155		170							
7/8	±1-1/2% of L with a minimum of ± .010 in.	156		171		184					
1		157		172		185		197			
1-1/4		158		173		186		198			
1-1/2		159		174		187		199			
1-3/4		160		175		188		200			
2		161		176		189		201			
2-1/2	± 1/32	162		177		190		202			
3		163		178		191		203			

Ⓐ CANCELLED SEE NOTE

Note: For illustration, material, protective coating and other pertinent data, see sheet 1.

Ⓐ CANCELLED AFTER 29 January 1964 . Use MS51963.

INTERCHANGEABILITY RELATIONSHIP - MS51963 can replace the cancelled screws of the same size. The cancelled MS51019 cannot always replace MS51963 and should be used until existing stock is depleted. Then use only the superseding MS screws of the same size for design and replacement.

APPROVED 7 MAY 58 REVISED Ⓐ FOR CHANGES SEE SHEETS 1 AND 2.

NOTICE—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

This military standard is approved by the Department of Defense and is mandatory on all activities. Selection for all new engineering and design applications and for repetitive use shall be made from this document.

P.A. WC Other Cust SHIPS ASD	TITLE SETSCREW, HEXAGON SOCKET, CUP POINT, ALLOY STEEL, PHOSPHATE COATED, NC-3A AND UNC-3A	MILITARY STANDARD MS51019	
PROCUREMENT SPECIFICATION FF-S-103	SUPERSEDES:	SHEET 2 OF	

DD FORM 1 SEP 57 672-1 (Cont'd) PREVIOUS EDITION OF THIS FORM IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23 17:53Z
Check the source to verify that this is the current version before use.